

AKHILESH KUMAR MISHRA

PYTHON FULL STACK DEVELOPER

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EXPERIENCE

Software Development Engineer

CSC e-Governance Services India Ltd. (MeitY, Government of India)

Ministry of Electronics & Information Technology, Govt. of India

12/2023 – PRESENT | New Delhi, India

- Re-architected monolithic web services into scalable, asynchronous **FastAPI** and **Django** microservices, improving reliability for **2 million+ users**.
- Designed and scaled transaction orchestration services for AEPS, MATM, and VATM platforms using Django REST Framework, handling **500,000+ daily transactions**.
- Implemented secure API authentication using **JWT** and role-based access control (**RBAC**), including OTP verification, reducing authentication latency by 50%.
- Developed an asynchronous task queuing pipeline using **Celery** and **Redis**, reducing processing bottlenecks by 35% under peak traffic.
- Tuned **PostgreSQL** and **MySQL** database schemas using indexing and query partitioning, lowering transaction query latency by 45%.
- Constructed real-time transaction tracking APIs utilizing **FastAPI WebSockets** to stream live status updates in under 500ms.

Freelance Software Engineer

Workforce Telemetry & Operations

Real-Time Telemetry & Geofencing Platform

06/2023 – 11/2023 | Remote, India

- Developed a real-time workforce telemetry and geofencing system utilizing **FastAPI**, **SQLAlchemy** (asyncio), and **React.js**, serving **1,000+ active field agents**.
- Built an interactive live telemetry dashboard in **React** using Ant Design to visualize coordinates and speed, reducing UI load times by 30%.
- Implemented a high-performance, asynchronous ingestion API to capture GPS locations, processing **10,000+ concurrent GPS pings with 99.9% uptime**.

SELECTED PROJECTS

Facial Recognition Attendance Dashboard

Developed an automated student attendance tracking application in Python using OpenCV (Haar Cascade & LBPH) and MySQL, with custom Excel reports via Pandas.

STACK: Python / OpenCV / Tkinter / MySQL / Pandas

Himalayan Edges Website Development

Engineered an e-commerce platform using Django and React.js with PWA features, hosted on AWS, boosting page speeds and usage by 30%.

STACK: Python / Django / React.js / SQLite / AWS / PWA

Persistent Task Management Dashboard

Programmed a desktop utility for task planning, category filtering, and JSON data persistence, featuring clean MVC architecture patterns.

STACK: Python / Tkinter / JSON / MVC Patterns

Traveling Portal & Landing Page

A fully responsive, modern front-end landing page for a travel agency built with HTML, CSS, and Bootstrap layouts.

STACK: HTML5 / CSS3 / JavaScript / Bootstrap

SUMMARY

Software Development Engineer specializing in Python Full Stack development, with a track record of building secure, scalable backend architectures and web systems. Expert in Python, Django, Django REST Framework (DRF), FastAPI, Celery, and database design in PostgreSQL/MySQL. Proven experience in migrating monoliths to async microservices, designing event-driven pipelines using Celery/Redis, and building real-time dashboards via WebSockets. Optimizes database queries and scales backend infrastructures using Docker.

EDUCATION

Bachelor of Technology (B.Tech.)

Kanpur Institute of Technology

2019 - 2023 | Kanpur, UP

SKILLS

BACKEND

Python

Django

FastAPI

Celery

Django REST Framework

SQLAlchemy (asyncio)

Pandas

FRONTEND & APIS

React.js

JavaScript

HTML5

CSS3

Bootstrap

RESTful APIs

WebSockets

JWT Security

DATA & MESSAGING

PostgreSQL

MySQL

SQLite

Redis cache/broker

database schema design

query indexing

TOOLS & CLOUD

Git

GitHub

Docker

Postman

Linux environments

OpenCV

LBPH face recognition

CERTIFICATIONS

GeeksforGeeks MERN Stack Certification

Coding Ninjas DBMS Certification

Coding Ninjas DSA Certification

IIT Kanpur Prutor Cyber Security Certification

ACHIEVEMENTS

- Graduated with First Class Honors, maintaining top academic rank.
- Earned a 4-Star Gold Badge in SQL on HackerRank.